Build & Evaluate a Linear Regression Model

House Price Predictor — California Housing Dataset

Task 1 | AI/ML Internship — Maincrafts Technology | Dhanushya Somu

1. Objective

The goal of this task was to build and evaluate a simple Linear Regression model that predicts median house prices using the California Housing dataset. This project introduces the complete machine learning workflow — loading data, exploring it, preparing it, training a model, and evaluating how well it performs — using Python, pandas, and scikit-learn.

2. Dataset

The dataset used is the California Housing dataset, built into scikit-learn (*sklearn.datasets.fetch_california_housing*). It contains 20,640 records, each representing a California census block group from the 1990 U.S. Census, with 8 numeric features and one target variable (median house value, in units of $100,000). The dataset had no missing values, confirmed using *df.info()*.

Feature	Meaning
MedInc	Median income of the block group
HouseAge	Median age of houses in the block group
AveRooms / AveBedrms	Average rooms / bedrooms per household
Population / AveOccup	Population of the area / average household size
Latitude / Longitude	Geographic location of the block group
MedHouseVal (Target)	Median house value ($100,000s) — the value predicted

3. Exploratory Data Analysis (EDA)

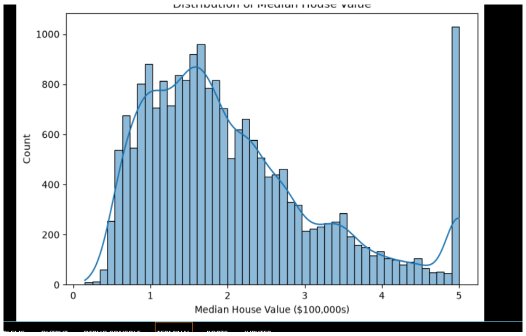
Before training any model, the dataset was explored to understand feature distributions and relationships with the target variable.

Correlation with target (MedHouseVal):

Feature	Correlation
MedInc	0.688
AveRooms	0.152
HouseAge	0.106
AveOccup	-0.024
Population	-0.025
Longitude	-0.046
AveBedrms	-0.047

Latitude	-0.144

Median income (MedInc) shows by far the strongest correlation with house price, confirming it as the most influential single feature. Other features show weak correlation individually, which is the reason all 8 features are used together in the regression model rather than relying on income alone.



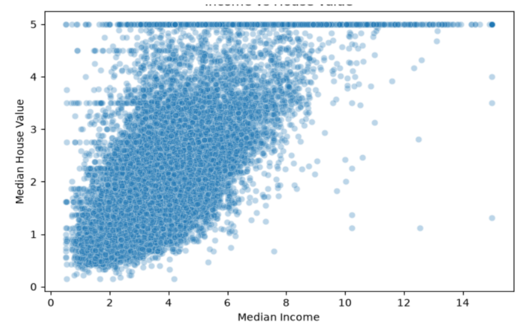


Figure 1 (left): Distribution of median house values — most houses fall between $100,000 and $300,000; the spike at 5.0 ($500,000) is a data-collection cap artifact, not real house density. Figure 2 (right): Median income vs. house value — a clear upward trend confirms income's strong positive correlation with price; the flat row of points at 5.0 reflects the same price cap.

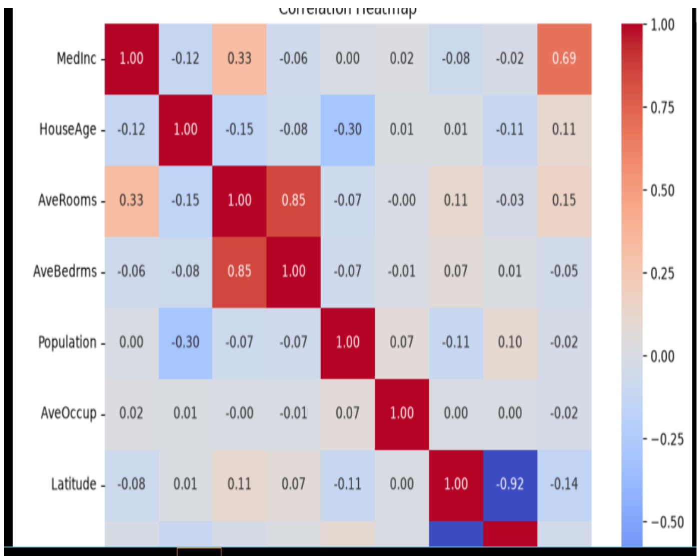


Figure 3: Correlation heatmap across all features. MedInc stands out as the strongest predictor of MedHouseVal (0.69). Latitude and Longitude are strongly negatively correlated with each other (-0.92), which is expected since they jointly describe the same geographic location.

4. Data Preparation & Model Training

The dataset was split into input features (X, 8 columns) and the target variable (y, MedHouseVal). It was then divided into a training set (80%, 16,512 records) and a test set (20%, 4,128 records) using *train_test_split* with *random_state=42* for reproducibility. A scikit-learn *LinearRegression* model was trained on the training set, then used to predict prices for the unseen test set.

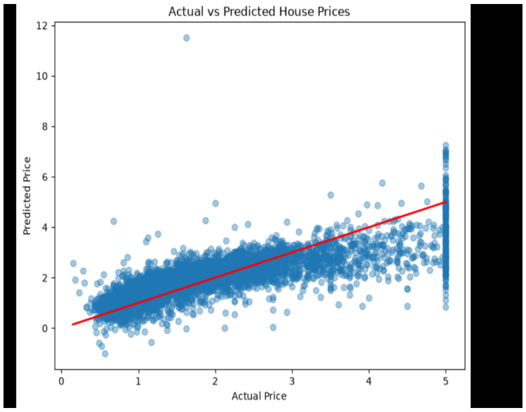
5. Model Evaluation

The model's predictions on the test set were evaluated using three standard regression metrics:

Metric	Value	Interpretation
MAE	0.533	On average, predictions are off by ~$53,300

RMSE	0.746	Similar to MAE, but penalizes larger errors more heavily
R^2 Score	0.576	The model explains ~57.6% of the variation in house prices

These results indicate a reasonable but imperfect model. RMSE being noticeably higher than MAE suggests a small number of predictions have relatively large errors, which pull the RMSE up. This is consistent with the visualizations below.



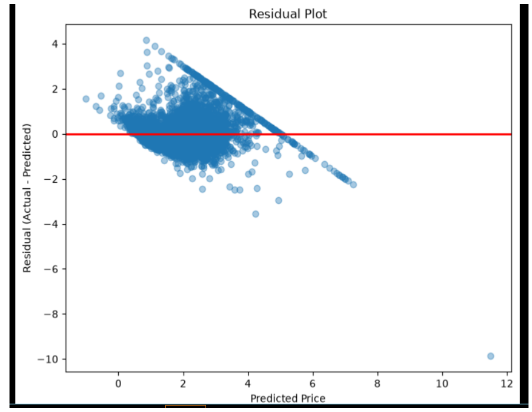


Figure 4 (left): Actual vs. predicted house prices — the red line marks perfect prediction; the vertical band at Actual = 5.0 corresponds to price-capped records. Figure 5 (right): Residual plot — most residuals cluster near zero; the diagonal streak reflects the same price cap, and one clear outlier appears near a predicted value of 11.5.

6. Limitations & Ideas for Improvement

- **Price cap distortion:** House values above $500,000 were capped during data collection, distorting predictions and error metrics for high-value areas. Removing or separately modeling these capped records could improve accuracy.
- **Non-linear relationships:** Linear Regression assumes straight-line relationships between features and price. Trying models such as Random Forest Regressor or Gradient Boosting could capture more complex patterns and likely improve R^2.
- **Feature engineering:** Creating new features (e.g., rooms per household, or combining latitude/longitude into distance-to-coast or distance-to-city) may improve predictive power, since location clearly matters but is currently only captured as raw coordinates.
- **Outlier handling:** A small number of predictions were far off (e.g., the outlier in Figure 5). Investigating and potentially removing or capping extreme outliers could improve overall model stability.

7. Conclusion

A Linear Regression model was successfully built and evaluated on the California Housing dataset, achieving an R^2 of 0.576, meaning it explains roughly 58% of the variation in house prices using 8 features. Median income was found to be the strongest single predictor. While the model performs reasonably well for a first baseline, the identified limitations — particularly the price cap and the linear-only assumption — present clear directions for improving accuracy in future iterations.